



# Surds

## SEE ALSO

◀ 36–39 Powers and roots

Fractions 48–55 ▶

A SURD IS A SQUARE ROOT THAT CANNOT BE WRITTEN AS A WHOLE NUMBER. IT HAS AN INFINITE NUMBER OF DIGITS AFTER THE DECIMAL POINT.

## Introducing surds

Some square roots are whole numbers and are easy to write. But some are irrational numbers—numbers that go on forever after the decimal point. These numbers cannot be written out in full, so the most accurate way to express them is as square roots.

$$\sqrt{5} = 2.2360679774\dots$$

irrational number

### △ Surd

The square root of 5 is an irrational number—it goes on forever. It cannot accurately be written out in full, so it is most simply expressed as the surd  $\sqrt{5}$ .

$$\sqrt{4} = 2$$

rational number

### △ Not a surd

The square root of 4 is not a surd. It is the number 2, a whole, or rational, number.

## Simplifying surds

Some surds can be made simpler by taking out factors that can be written as whole numbers. A few simple rules can help with this.

### ▷ Square roots

A square root is the number that, when multiplied by itself, equals the number inside the root.

$$\sqrt{a} \times \sqrt{a} = a$$

$$\sqrt{3} \times \sqrt{3} = 3$$

multiply the surd by itself to get the number inside the square root

### ▷ Multiplying roots

Multiplying two numbers together and taking the square root of the result equals the same answer as taking the square roots of the two numbers and multiplying them together.

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$$

$$\sqrt{48} = \sqrt{16 \times 3} = \sqrt{16} \times \sqrt{3} = 4 \times \sqrt{3}$$

look for factors that are square numbers

48 can be written as  $16 \times 3$

the square root of 16 is a whole number

the square root of 3 is an irrational number, so it stays in surd form

$\sqrt{16} = 4$ , so this can be written as  $4 \times \sqrt{3}$